G11 Physics (Group 2)

Instructor

Tashi Wangchuk
Email: tashiwangchuk@academy.bt
Phone: +975 17 494 290

Office Hours

I do not have formal office hours, but I am usually in or around the Dzong courtyard during weekdays. Please feel free to speak with me or ask questions anytime you see me. I am also easily reached via email; if you have questions, send them when you are on your laptop and I will get back to you, ideally in person.

Course Description

This course moves beyond memorizing facts to explore the structural unity of our universe through conservation laws and symmetry. The contents are based on an initial synthesis of the Grade 11 Bhutan Baccalaureate curriculum, the current Grade 12 BHSEC curriculum, and a Grade 11 Cambridge-aligned curriculum. While we follow a strong Cambridge-aligned foundation, we go further to prepare you for the world of professional science. You will learn to treat mathematics as the language of physical reality and develop the Research Skills needed to design original experiments and communicate your findings. Our goal is to help you bridge the gap between being a student of physics and thinking like a scientist.

Learning component	**# lessons**	**Approx. weeks**
Atoms in Motion	2	0.5
Dimensional Analysis and Vectors	5	1
Motion	10	2
Newton's Laws of Dynamics	18	4
Momentum and Angular Momentum	10	2
Energy	10	2
Research Skills	6	1–2
Experiment 1	8	2
The Theory of Gravitation	8	2
Electrostatics	10	2–3
Electric Circuits	8	2
Magnetostatics	10	2–3
Experiment 2	8	2
Optics	12	3
Modern Physics (Quantum Mechanics Preview)	10	2–3
Brownian Motion (Statistical Mechanics Preview)	6	1–2
Total	**146**	

Detailed Contents

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1. Atoms in Motion (What is our overall picture of the world?)

1.1 Introduction to physics
1.2 Matter is made of atoms
1.3 Atomic processes (atomic hypothesis)

2. Dimensional Analysis and Vectors

2.1 Dimensional analysis
2.2 Vector operations
2.3 Vector algebra: component form
2.4 Position, displacement, and separation vectors

3. Motion

3.1 Description of motion
3.2 Speed as a derivative (introducing differentiation)
3.3 Speed and velocity $\left(s = \frac{ds}{dt},\ \vec{v} = \frac{d\vec{r}}{dt}\right)$
3.4 Distance as an integral (introducing integration)
3.5 Acceleration
3.6 Describing circular motion (as in sec. 5.1 of the Cambridge-aligned curriculum)

4. Newton's Laws of Dynamics

4.1 Newton's first law of motion
4.2 Newton's second law of motion
- 4.2.1 Components of velocity, acceleration, and force
- 4.2.2 What is force?
- 4.2.3 Rephrase Newton's 2^{nd} law in terms of a particle's momentum (introducing momentum simultaneously)
- 4.2.4 Impulse–momentum theorem
- 4.2.5 Constant force $\rightarrow$ constant acceleration (recover kinematic equations)

4.3 Newton's third law of motion
4.4 Experience in an elevator (Newton's 1^{st} and 2^{nd} laws in action)
- 4.4.1 How does apparent weight change in an accelerating elevator?
- 4.4.2 Introduce the principle of equivalence

5. Research Skills

5.1 Developing an appreciation of research
- 5.1.1 The research cycle
- 5.1.2 Finding scientific information (internet, library, journals, databases, citation indices)

5.2 Writing scientific documents
- 5.2.1 Structure of scientific papers
- 5.2.2 Dictum of tabulation
- 5.2.3 Canon of graphing

5.3 Presenting scientific information
- 5.3.1 Oral presentations (PowerPoint or similar)
- 5.3.2 Sizing figures and text appropriately
- 5.3.3 Balancing text with images
- 5.3.4 Proper contrast and audience awareness

5.4 Measurement and uncertainty
- 5.4.1 Instrumental limitations and repeatability
- 5.4.2 Mean, standard deviation, and standard deviation of the mean
- 5.4.3 Normal distribution
- 5.4.4 Propagation of uncertainty (fundamentals)

5.5 Processing data
- 5.5.1 Spreadsheets (Excel)
- 5.5.2 Simple programming
- 5.5.3 Data fitting and plotting

6. Experiment 1

6.1 Experimental investigation (e.g., speed of sound via time-of-flight)
- 6.1.1 Guided instruction without detailed step-by-step procedures
- 6.1.2 Design and refinement of experimental procedure
- 6.1.3 Data collection and uncertainty estimation
- 6.1.4 Data analysis and interpretation

6.2 Scientific reporting
- 6.2.1 Structured scientific report
- 6.2.2 Draft with formative feedback
- 6.2.3 Revision and final submission

7. Momentum and Angular Momentum

7.1 Conservation of momentum
- 7.1.1 Rocket propulsion

7.2 The center of mass

7.3 Angular momentum for a single particle (torque as its rate of change)
- 7.3.1 Kepler's second law as a beautiful consequence

8. Energy

8.1 Kinetic energy and work (work–KE theorem)
8.2 Potential energy and conservative forces
  8.2.1 Conditions for a force to be conservative
  8.2.2 Principle of conservation of energy for one particle
  8.2.3 Nonconservative forces
8.3 Collision in 2D

9. The Theory of Gravitation

9.1 Planetary motions
9.2 Kepler's laws (second law previously derived from conservation of angular momentum in 7.3.1)
9.3 Development of dynamics
9.4 Newton's law of gravitation
9.5 Universal gravitation
9.6 Cavendish's experiment
9.7 What is gravity?
9.8 Gravity and relativity (brief touch)
  9.8.1 Spacetime curvature demonstration (fabric analogy)

10. Electrostatics

10.1 Electric charge (source and test charge)
10.2 Quarks (fractional electric charge)
10.3 Coulomb's law
10.4 The electric field
10.5 Electric potential and potential difference

11. Electric Circuits

11.1 Electric current
11.2 Potential difference and EMF
11.3 Ohm's law
11.4 Resistance and resistivity
11.5 Electrical power and energy
11.6 Internal resistance
11.7 Resistor combinations

12. Magnetostatics

12.1 Magnet
12.2 Atomic/molecular theory of magnetisation
12.3 Magnetic field
12.4 Terrestrial magnetism
12.5 Atom as a magnetic dipole
12.6 Magnetic flux and magnetic flux density
12.7 Projectiles and charged particles
- 12.7.1 A curved trajectory
- 12.7.2 Motion of charged particles in uniform magnetic field
- 12.7.3 Motion of charged particles in electric and magnetic fields (tentative)

13. Experiment 2

13.1 Experimental investigation (different from Experiment 1)
- 13.1.1 Learners will complete the full research cycle as in Experiment 1
- 13.1.2 Written scientific report
- 13.1.3 Oral presentation of findings (extra)

14. Optics

14.1 Wave Optics
- 14.1.1 One-dimensional waves (introducing transverse and longitudinal waves, and the 1D wavefunction $\psi(x,t) = f(x \pm vt)$)
- 14.1.2 The wave equation
- 14.1.3 Plane electromagnetic waves and their velocity $v = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$ (connecting to the speed of light)
- 14.1.4 Rayleigh scattering

14.2 Ray Optics
- 14.2.1 Reflection and refraction
- 14.2.2 Total internal reflection
- 14.2.3 Reflection of light from a spherical mirror
- 14.2.4 Fermat's principle of least time
- 14.2.5 Applications of Fermat's principle

15. Modern Physics (tentative)

15.1 Atomic mechanics (basics of particle physics)
15.2 An experiment with bullets
15.3 An experiment with waves
15.4 An experiment with electrons
15.5 The interference of electron waves
15.6 Watching the electrons
15.7 First principles of quantum mechanics
15.8 The uncertainty principle

16. Brownian Motion (tentative)

16.1 Background (relate to statistical mechanics)
16.2 Random walks
16.3 Dynamical equations (Stoke–Einstein–Sutherland formula)
16.4 Experimental observation

Textbooks

1. *Physics for Class 11 Textbook: Aligned to Cambridge Curriculum Standards*, by Centre for School Curriculum Development, Department of School Education, Ministry of Education and Skills Development.

2. *The Feynman Lectures on Physics (Volumes 1 and 2)*, by Richard Feynman (`https://www.feynmanlectures.caltech.edu/`)

3. *Concepts of Physics (Part 1 and 2)*, by H.C. Verma

4. *Introduction to Electrodynamics*, by David J. Griffiths (5th Edition)

5. *Fundamentals of Physics*, by Jearl Walker (10th Edition)

6. *Classical Mechanics*, by John R. Taylor

7. *Optics*, by Eugene Hecht (5th Edition)

8. *Introduction to Quantum Mechanics*, by David J. Griffiths and Darrel F. Schroeter (3rd Edition)

9. *An Introduction to Error Analysis*, by John R. Taylor (2nd Edition). This is an excellent resource for anyone continuing in Science; it is strongly recommended that you acquire a copy for your personal library.

Course Work and Grading

Component	Weight	Description
Assignments	15%	Standard written problem sets (10 – 15 total).
Experiment 1	15%	Full research cycle on a specific topic.
Experiment 2	20%	Same as Experiment 1, with an added oral presentation
5 Reviews	25%	3 Individual reviews and 2 cross-pollination reviews.
Few in-class tests	20%	Comprehensive assessment (divided into standard and take-home).
Presentation	5%	Presentation with feedback to other students

Labs

Labs are the heartbeat of this course. You will complete two major experimental investigations and several smaller measurements.

- **Guided autonomy:** For major experiments, you will receive instruction on goals, but we avoid "cookbook" step-by-step procedures to allow for your own thinking.
- **Lab instruments:** Some experiments may be conducted using provided kits or mobile (like smartphone sensors for time-of-flight measurements).
- **Lab records:** All experimental data and observations must be recorded in your notebook. These records are critical and will be reviewed periodically.
- **Presentation:** For your 2nd experiment you will prepare a presentation to give in the last days of the class. You will be expected to attend, ask questions and provide feedback to the other students giving presentations.